

MIDTERM EXAMINATION
Fall 2009
MTH101- Calculus And Analytical Geometry (Session - 5)

Ref No: 1087754
Time: 60 min

Question No: 1 (Marks: 1) - Please choose one

_____ If
f is a twice differentiable function at a stationary point x_0 and $f''(x_0) < 0$ then f has
relative At x_0

- ▶ Minima
- ▶ **Maxima**
- ▶ None of these

Note:Maxima(If Maxima refers to local maximum)

Question No: 2 (Marks: 1) - Please choose one

_____ A
line $x = x_0$ is called ----- for the graph of a function f if
 $f(x) \rightarrow +\infty$ or $f(x) \rightarrow -\infty$ as x approaches x_0 from the right or from the left

- ▶ Horizontal asymptotes
- ▶ None of these
- ▶ **Vertical asymptotes**

Question No: 3 (Marks: 1) - Please choose one

_____ A
line $y = y_0$ is called a for the graph f if
 $\lim_{x \rightarrow +\infty} f(x) = y_0$ or $\lim_{x \rightarrow -\infty} f(x) = y_0$

- ▶ Vertical asymptotes
- ▶ Horizontal asymptotes
- ▶ **None of these**

Question No: 4 (Marks: 1) - Please choose one

_____ According to Power-Rule of differentiation, if $f(x) = x^n$ where n is a real number, then

$$\frac{d}{dx}[x^n] =$$

- ▶ x^{n-1}
- ▶ $n x^{n-1}$
- ▶ $n x^{n+1}$
- ▶ $(n-1)x^{n+1}$

Question No: 5 (Marks: 1) - Please choose one

If

$$y = \frac{1}{1-x} \quad \frac{dy}{dx} =$$

then

- ▶ 1
- ▶ -1
- ▶ $\frac{1}{(1-x)^2}$
- ▶ $\frac{-1}{(1-x)^2}$

Question No: 6 (Marks: 1) - Please choose one

If

$$xy = 4 \quad \frac{dy}{dx} =$$

then

- ▶ 0
- ▶ $\frac{-1}{x^2}$
- ▶ $\frac{4}{x^2}$
- ▶ $\frac{-4}{x^2}$

Question No: 7 (Marks: 1) - Please choose one

If

$$2x - y = -3 \quad \frac{dy}{dx} =$$

then

- ▶ 2

- ☒ -2
- ☐ 0
- ☐ -3

Question No: 8 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\sec x] = \underline{\hspace{2cm}}$$

- ☐ $\frac{1}{1+\sin^2 x}$
- ☐ $\frac{-\sin x}{1+\sin^2 x}$
- ☐ $\frac{1}{1-\sin^2 x}$
- ☒ $\frac{\sin x}{1-\sin^2 x}$

Question No: 9 (Marks: 1) - Please choose one

$$= \underline{\hspace{2cm}} \quad 30^\circ$$

- ☐ $\frac{\pi}{3}$
- ☐ $\frac{\pi}{4}$
- ☒ $\frac{\pi}{6}$
- ☐ $\frac{\pi}{2}$

Question No: 10 (Marks: 1) - Please choose one

Consider a function $h(x)$ and a constant c then

$$\frac{d}{dx}((c) \{h(x)\}) = \underline{\hspace{2cm}}$$

- ▶ 0
- ▶ $\frac{d}{dx}(h(x))$
- ▶ $\frac{d}{dx}(h(cx))$
- ▶ $c \frac{d}{dx}(h(x))$

Question No: 11 (Marks: 1) - Please choose one

$$\frac{d}{dx}[\operatorname{cosec} x] = \underline{\hspace{2cm}}$$

- ▶ $\frac{1}{1 + \cos^2 x}$
- ▶ $\frac{-\cos x}{1 - \cos^2 x}$
- ▶ $\frac{-\cos x}{1 - \cos^2 x}$
- ▶ $\frac{1}{1 - \cos^2 x}$

Question No: 12 (Marks: 1) - Please choose one

Chain rule is a rule for differentiating _____ of functions.

- ▶ **Product**
- ▶ Sum
- ▶ Difference
- ▶ Composition

Question No: 13 (Marks: 1) - Please choose one

_____ The
$$\frac{d}{dx}[x^n] = nx^{n-1}$$

power rule, _____ holds if n is _____

- ▶ **An integer**
- ▶ A rational number
- ▶ An irrational number
- ▶ All of the above

Question No: 14 (Marks: 1) - Please choose one

_____ Let
a function f be defined on an interval, and let x_1 and x_2 denote points in that interval. If $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ then which of the following statement is correct?

- ▶ **f is an increasing function.**
- ▶ f is a decreasing function.
- ▶ f is a constant function.

Question No: 15 (Marks: 1) - Please choose one

_____ If
 $f''(x) > 0$ on an open interval (a,b), then which of the following statement is correct?

- ▶ **f is concave up on (a, b).**
- ▶ f is concave down on (a, b).
- ▶ f is linear on (a, b).

Question No: 16 (Marks: 1) - Please choose one

_____ If
 $f''(x) < 0$ on an open interval (a,b) then which of the following statement is correct?

- ▶ f is concave up on (a, b).
- ▶ **f is concave down on (a, b)**
- ▶ f is linear on (a, b).

Question No: 17 (Marks: 1) - Please choose one

_____ If $x > 0$ then $\frac{d}{dx}[\ln x] =$ _____

- ▶ 1
- ▶ x

☒ $\frac{1}{x}$

☐ $\ln \frac{1}{x}$

Question No: 18 (Marks: 1) - Please choose one

If $b > 0$ then $\frac{d}{dx}[b^x] =$ _____

☐ 0

☐ xb^{x-1}

☐ $\ln b$

☒ $b^x \ln b$

Question No: 19 (Marks: 1) - Please choose one

$y = (x^3 + 2x)^{37}$. Which of the following is correct?

☐ $\frac{dy}{dx} = (37)(x^3 + 2x)^{36}$



☐ $\frac{dy}{dx} = 111x^2(x^3 + 2x)^{36}$



☐ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{36}$



☒ $\frac{dy}{dx} = (111x^2 + 74)(x^3 + 2x)^{38}$



Question No: 20 (Marks: 1) - Please choose one

What is the base of natural logarithm?

☒ 2.71

☐ 10

Let

- ▶ 5
- ▶ Any real number

Question No: 21 (Marks: 1) - Please choose one

Let x_0 be critical points of the function f . Those critical points for which $f'(x_0) = 0$ are called _____ of f

- ▶ Local points
- ▶ End points
- ▶ **Stationary points**

Question No: 22 (Marks: 1) - Please choose one

$\log_b a^r = \underline{\hspace{2cm}}$

- ▶ $a \log_b r$
- ▶ $r \log_b a$
- ▶ $\frac{\log_b a}{\log_b r}$
- ▶ $\log_b a + \log_b r$

Question No: 23 (Marks: 1) - Please choose one

$\log_b \frac{1}{c} = \underline{\hspace{2cm}}$

- ▶ $\log_b c$
- ▶ $1 - \log_b c$
- ▶ $-\log_b c$
- ▶ $1 + \log_b c$

Question No: 24 (Marks: 1) - Please choose one

$$\log_b \frac{1}{t} = \underline{\hspace{2cm}}$$

- ▶ $\log_b t$
- ▶ $1 - \log_b t$
- ▶ $1 + \log_b t$
- ▶ $-\log_b t$

Question No: 25 (Marks: 3)

If $f(x) = x^4 - 8x^2$, determine all relative extrema for the function. Using First Derivative Test.

Solution:

$$f = x^4 - 8x^2$$

$$f' = 4x^3 - 16x$$

$$f' = 0$$

$$4x^3 - 16x = 0$$

$$x(4x^2 - 16) = 0$$

$$x = 0$$

$$4x^2 - 16 = 0$$

$$x^2 = \frac{16}{4}$$

$$x^2 = 4$$

$$x = \pm 2$$

Relative extrema $(0, \pm 2)$Ans

Question No: 26 (Marks: 5)

Differentiate $y = x^{-2}(4 + 3x^{-3})$

Solution:

$$y = 4x^{-2} + 3x^{-3} \cdot x^{-2}$$

$$= 4x^{-2} + 3x^{-5}$$

$$\frac{dy}{dx} = 4 \frac{d}{dx}(x^{-2}) + 3 \frac{d}{dx}(x^{-5})$$

$$= 4(-2)x^{-2-1} + 3 \frac{d}{dx} - 5x^{-5-1}$$

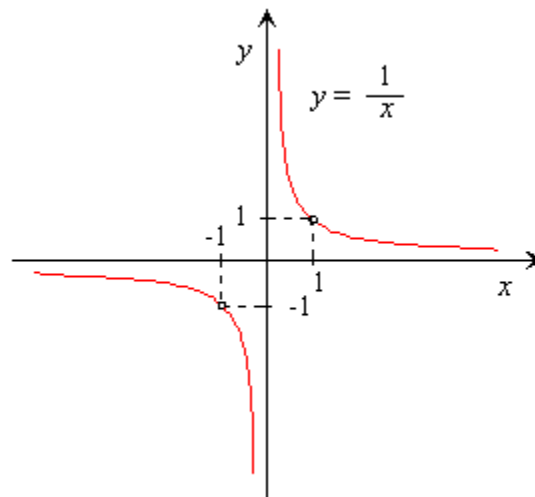
$$= -8x^{-3} + (-15x^{-6})$$

$$\frac{dy}{dx} = -8x^{-3} - 15x^{-6} \dots \dots \text{Ans}$$

Question No: 27 (Marks: 10)

$$f(x) = \frac{1}{x}$$

Determine the intervals in which the graph of the function is concave upward or downward.



Solution:

$$f(x) = \frac{1}{x}$$

$$f'(x) = -\frac{1}{x^2}$$

X	1	2	3	4
F(x)	-1	-0.25	-0.11	-0.625

Conclusion:

$f'(x)$ is increasing when x is from $(0, \infty)$

so,

It is concave up.

$f'(x)$ is decreasing when x is from $(-\infty, 0)$

so,

It is concave down